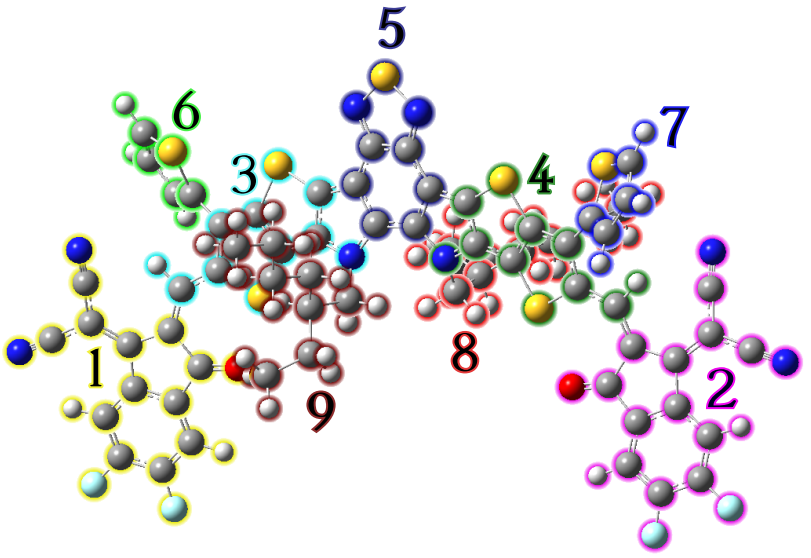
a)



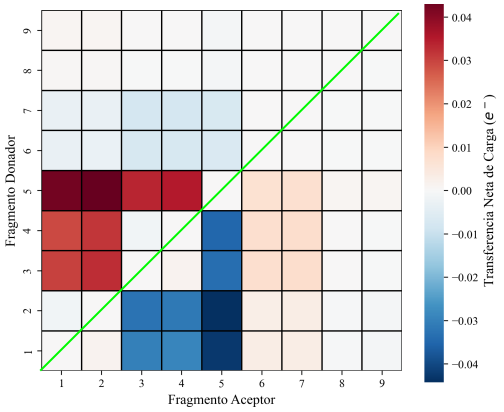
b)

HOMO→LUMO 100%

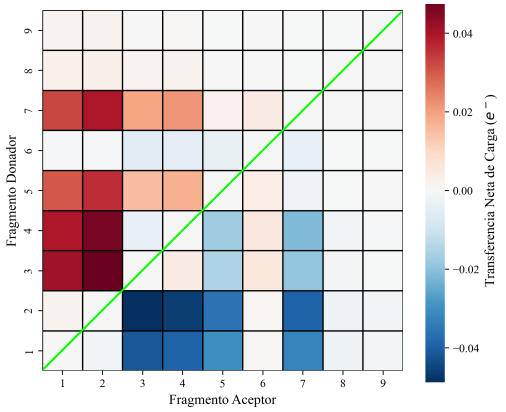
HOMO→LUMO 59%
HOMO-1→LUMO+1 39%

HOMO→LUMO 95%

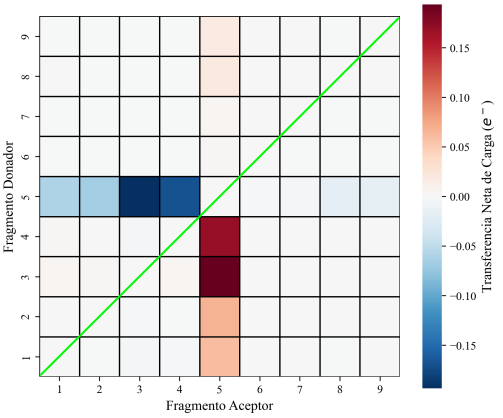
1 EE



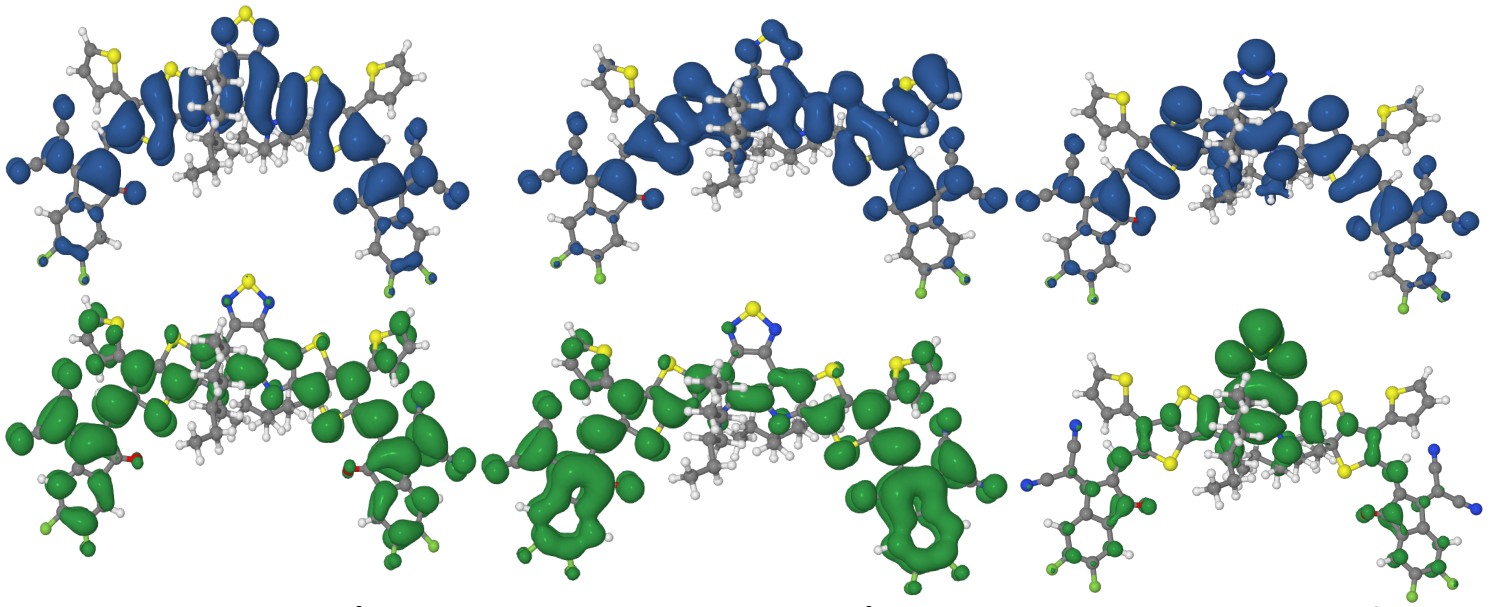
6 EE



37 EE



c)



$D_{index} = 1.284$ Å

$D_{index} = 2.424$ Å

$D_{index} = 2.498$ Å